# Physics 9702 Paper 4 · Topic 25

## Astronomy and cosmology

The seven hardest A Level Structured Questions on this topic from the 2020 to 2026 papers, hardest first, followed by their mark schemes.

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**1 9702/41 May/June 2022, Question 10** (9 marks) page 2

*Examiner report:* (ii) Some candidates were confused here and thought star B had a smaller wavelength because its maximum intensity peak was lower. (ii) Only the strongest candidates gained credit.

**2 9702/41 October/November 2025, Question 9** (7 marks) page 4

*Examiner report:* (b) Many candidates wrote at least one answer that, although factually correct, was not a conclusion that could be drawn from the data.

**3 9702/42 February/March 2024, Question 10** (10 marks) page 6

*Examiner report:* (ii) Most candidates found this a difficult question, but many of the stronger candidates were able to calculate the correct answer.

**4 9702/42 February/March 2022, Question 12** (7 marks) page 8

*Examiner report:* Some candidates omitted the key word “total” and others confused power with energy or intensity. (c) This calculation was competed well, but a significant minority of candidates used the radiant flux intensity at the Earth in (b) rather than the luminosity of the Sun.

**5 9702/42 October/November 2023, Question 10** (8 marks) page 10

*Examiner report:* (a) Most candidates knew that Wien’s displacement law involves a wavelength being inversely proportional to a temperature, but only the strongest candidates were able to correctly identify what, in the context of Wien’s law, is meant by wavelength and temperature. (b) (i) This question was generally well answered, although a significant minority of candidates attempted to describe what standard candles are used for rather than stating what one is.

**6 9702/42 May/June 2024, Question 8** (12 marks) page 11

*Examiner report:* Many candidates struggled to articulate this using the correct technical language, but the more able candidates were usually able to achieve at least partial credit. Only the strongest candidates appreciated how the photon energy needs to be combined with the –3.40 eV energy level to get to the answer of –0.85 eV.

**7 9702/42 May/June 2022, Question 9** (7 marks) page 13

*Examiner report:* Many candidates described decreased wavelengths or even wavelengths that are continuously increasing. Many confused the Universe with the stars being observed and tried to suggest that the Universe is moving away from the Earth, or suggested that the observation gives evidence that stars are moving away from each other.


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Questions were chosen using the Cambridge examiner reports (Principal Examiner Report for Teachers): parts the reports describe as challenging, poorly answered, or answered fully only by the strongest candidates. May/June 2020 and May/June 2026 have no report, so questions from those series were judged directly. Variant 43 repeats variant 41 in every series, so only 41 is cited.

**10** **(a)** State Wien's displacement law.

.....

..... [1]

**(b)** Fig. 10.1 shows the wavelength distributions of electromagnetic radiation emitted by two stars A and B.

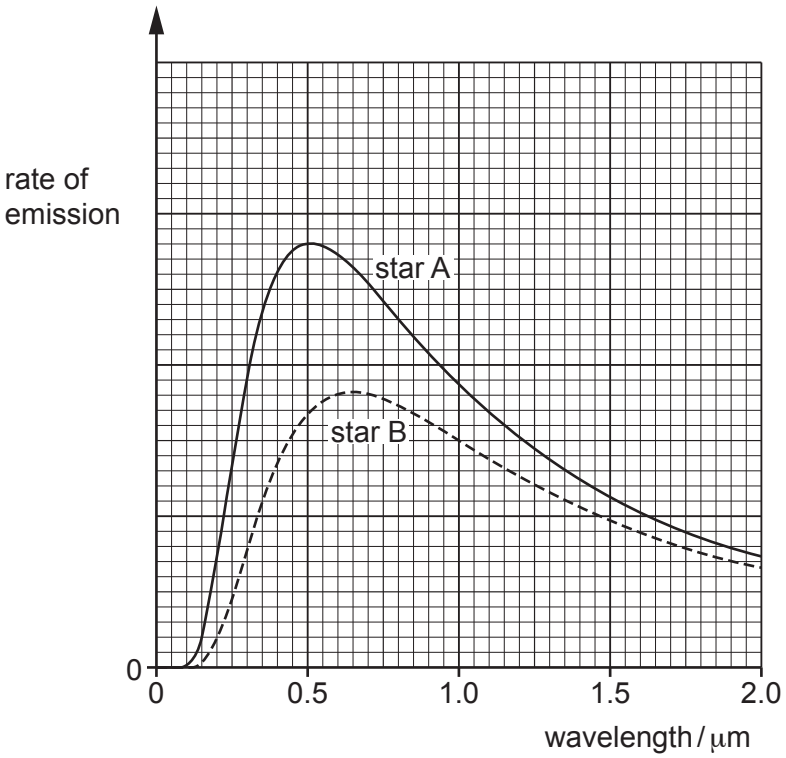



**Fig. 10.1**

The surface temperature of star A is known to be 5800 K.

**(i)** Determine the surface temperature of star B.

surface temperature = ..... K [2]

**(ii)** Star B appears less bright than star A when viewed from the Earth.

Use Fig. 10.1 to suggest, with a reason, how else the physical appearance of star B compares with that of star A.

.....

.....

..... [2]

**(c)** The lines in Fig. 10.1 have been corrected for redshift.

**(i)** State what is meant by redshift.

.....

.....

..... [2]

**(ii)** Explain how cosmologists are able to determine that light from a distant star has undergone redshift.

.....

.....

..... [2]

[Total: 9]

**9** **(a)** State Wien's displacement law.

.....

.....

..... [2]

**(b)** Fig. 9.1 shows the variation with $d^{-2}$ of the radiant flux intensity $F$ observed from a star X, where $d$ is the distance of the observer from the star. Fig. 9.2 shows the variation with wavelength $\lambda$ of the rates of emission $P$ of radiation by star X and the Sun.

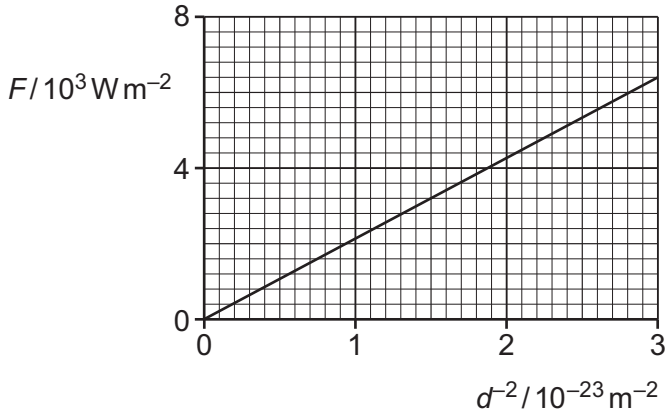



**Fig. 9.1**

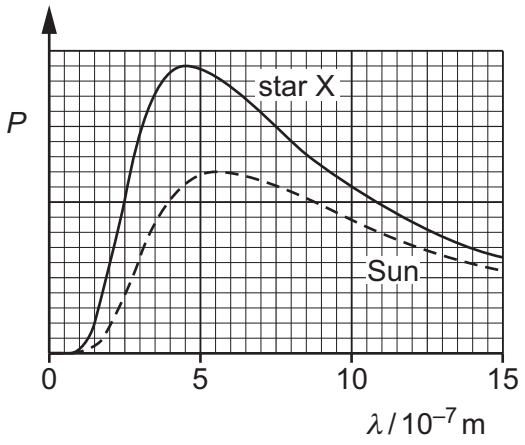



**Fig. 9.2**

The surface temperature of the Sun is 5770 K.

State **three** conclusions about star X that can be drawn from this data. The conclusions may be qualitative or quantitative. Use the space for any working.

1 .....

.....

2 .....

.....

3 .....

..... [3]

**(c)** Star X is in a galaxy that is moving away from the Earth.

Suggest, with a reason, how the line for star X in Fig. 9.2 would appear differently if it had been obtained from data measured on the Earth.

.....

.....

..... [2]

[Total: 7]

**10** **(a)** The Sun has a surface temperature of 5780 K. The luminosity of the Sun is $3.85 \times 10^{26}$ W.

**(i)** Calculate the radius of the Sun.

radius = ..... m [2]

**(ii)** The Earth is a distance of $1.50 \times 10^{11}$ m from the Sun.

Calculate the radiant flux intensity $F$ of the radiation from the Sun at a distance of $1.50 \times 10^{11}$ m. Give a unit with your answer.

$F$ = ..... unit ..... [2]

**(iii)** The variation with wavelength of the intensity of radiation emitted from the Sun is shown in Fig. 10.1.

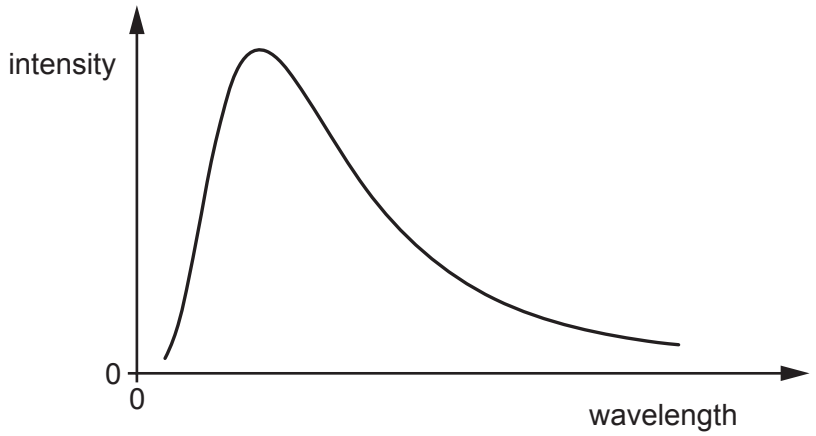



**Fig. 10.1**

Another star has the same radius as the Sun but has a lower surface temperature.

On Fig. 10.1, sketch a line to show the variation with wavelength of the intensity of the radiation emitted for this star. [2]

**(b)** A galaxy in the constellation Corona Borealis is moving away from the Earth.

**(i)** The visible emission spectrum for the Sun is shown in Fig. 10.2.

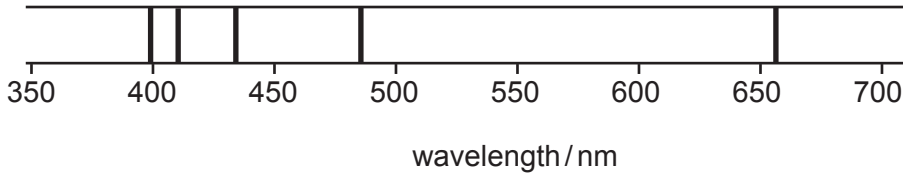



**Fig. 10.2**

The lines are at wavelengths of 397 nm, 410 nm, 434 nm, 486 nm and 656 nm. The compositions of the Sun and a star in the Corona Borealis galaxy are similar.

On Fig. 10.3, sketch the emission spectrum for the star in the Corona Borealis galaxy as observed from the Earth. No calculations are required.

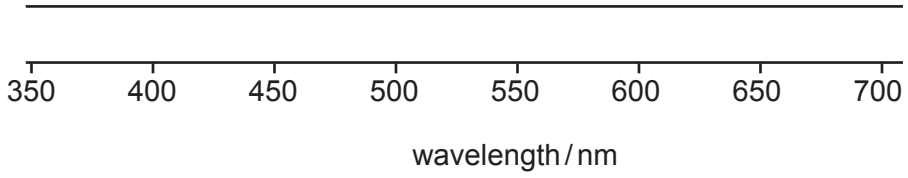



**Fig. 10.3**

[1]

**(ii)** The galaxy in Corona Borealis is moving away from the Earth at a speed of 21 400 km $s^{-1}$.

Use information from **(b)(i)** to calculate, in nm, the observed wavelength of the lowest visible energy emission for the star in the Corona Borealis galaxy.

wavelength = ..... nm [2]

**(iii)** The wavelength in **(b)(ii)** is used to calculate a value for the surface temperature of the star in the Corona Borealis galaxy. The calculation does not give an accurate value.

State and explain whether this value of temperature is too high or too low.

.....

.....

..... [1]

[Total: 10]

**12** **(a)** State what is meant by luminosity of a star.

.....

..... [1]

**(b)** The luminosity of the Sun is $3.83 \times 10^{26}$ W. The distance between the Earth and the Sun is $1.51 \times 10^{11}$ m.

Calculate the radiant flux intensity $F$ of the Sun at the Earth. Give a unit with your answer.

$F$ = ..... unit ..... [2]

**(c)** Use data from **(b)** to calculate the mass that is converted into energy every second in the Sun.

mass = ..... kg [1]

**(d)** The radius of the Sun is $6.96 \times 10^{8}$ m.

Show that the temperature $T$ of the surface of the Sun is 5770 K.

[1]

**(e)** The wavelength $\lambda_{max}$ of light for which the maximum rate of emission occurs from the Sun is $5.00 \times 10^{-7}$ m.
The temperature of the surface of the star Sirius is 9940 K.

Use information from **(d)** to determine the wavelength of light for which the maximum rate of emission occurs from Sirius.

wavelength = ..... m [2]

[Total: 7]

**10** **(a)** State Wien's displacement law. Identify any symbols that you use.

.....

.....

..... [2]

**(b)** A cosmology student observes the electromagnetic radiation received from a star in a galaxy. The student uses Wien's law to estimate the surface temperature of the star, a standard candle to estimate the distance to the galaxy, and the Stefan–Boltzmann law to estimate the radius of the star.

The student observes that the radiation from the star is redshifted.

**(i)** State what is meant by a standard candle.

..... [1]

**(ii)** State the reason why the radiation from the star is redshifted.

..... [1]

**(iii)** The true values of the quantities observed or estimated are those that are corrected to allow for redshift. However, the student does not correct for redshift.

By placing one tick (✓) in each row, complete Table 10.1 to indicate how the observations and estimates made by the student compare with the true values.

**Table 10.1**

| | student's uncorrected value | | |
|---|---|---|---|
| | too low | the same | too high |
| wavelength of radiation | | | |
| surface temperature of star | | | |
| distance to star | | | |
| radius of star | | | |

[4]

[Total: 8]

**8** Fig. 8.1 shows part of the emission spectrum of visible radiation emitted by hydrogen gas in a star in a distant galaxy.

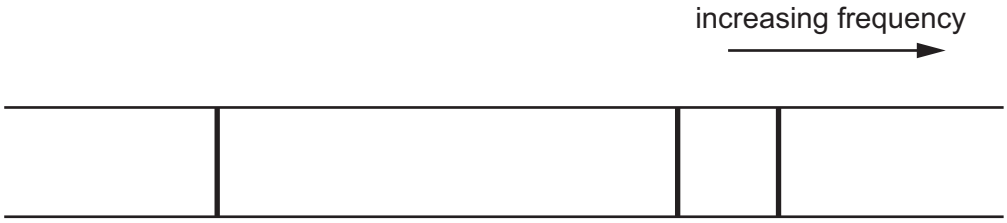



**Fig. 8.1**

The galaxy is moving away from the Earth at a speed of $6.2 \times 10^6\,\text{m s}^{-1}$.

**(a) (i)** Explain how the positions of the lines in the emission spectrum seen by an observer on the Earth differ from the positions shown in Fig. 8.1.

.....

.....

..... [2]

**(ii)** On Fig. 8.1, draw the three lines in possible positions in the spectrum seen by the observer. [2]

**(b)** The lines in Fig. 8.1 correspond to electron transitions down to the energy level –3.40 eV. One of the lines represents emitted radiation of wavelength 488 nm.

**(i)** Calculate the energy of a photon of this radiation.

photon energy = ..... J [2]

**(ii)** Determine the energy, in eV, of the energy level from which the electron transition originates to cause the emission of this radiation.

energy level = ..... eV [2]

**(iii)** Determine the wavelength, in nm, of this radiation as detected by the observer on the Earth.

wavelength = ..... nm [2]

**(c)** A value for the Hubble constant is $2.3 \times 10^{-18}\,s^{-1}$.

Determine the distance of the galaxy from the Earth.

distance = ..... m [2]

[Total: 12]

**9** **(a)** **(i)** State Hubble's law.

.....

.....

..... [2]

**(ii)** Explain how cosmologists use observations of emission spectra from stars in distant galaxies to determine that the Universe is expanding.

.....

.....

.....

..... [2]

**(b)** Explain how Hubble's law and the idea of the expanding Universe lead to the Big Bang theory of the origin of the Universe.

.....

.....

.....

..... [3]

[Total: 7]

| Question | Answer | Marks |
|---|---|---|
| 10(a) | wavelength of maximum intensity is inversely proportional to (thermodynamic) temperature | **B1** |
| 10(b)(i) | $\lambda_{MAX}$ = 0.50 μm for A **and** 0.65 μm for B | **C1** |
| | $T$ = 5800 × (0.50 / 0.65)<br>= 4500 K | **A1** |
| 10(b)(ii) | (star B has) greater peak / average wavelength | **B1** |
| | (star B looks) redder | **B1** |
| 10(c)(i) | apparent wavelength is greater<br>**or**<br>wavelength is greater than known value | **B1** |
| | (due to) movement of star away (from observer) | **B1** |
| 10(c)(ii) | by examining the (lines in the) spectrum (of light from the star) | **B1** |
| | and comparing with known spectrum | **B1** |

| Question | Answer | Marks |
|---|---|---|
| 9(a) | temperature inversely proportional to wavelength | **M1** |
| | temperature is thermodynamic temperature of surface of star **and** wavelength is the wavelength at which maximum emission rate from star occurs | **A1** |
| 9(b) | *Any three points from:*<br><br>• (surface) temperature of star X = 7000 K<br>**or**<br>star X has a higher temperature than the Sun<br><br>• star X has a higher luminosity than the Sun<br><br>• luminosity of star X = $2.7 \times 10^{27}$ W<br><br>• radius of star X = $1.3 \times 10^{9}$ m | **B3** |
| 9(c) | light (from star X) is redshifted | **B1** |
| | wavelength of peak emission rate would be greater (using observed data) | **B1** |

| Question | Answer | Marks |
|---|---|---|
| 10(a)(i) | $L = 4\pi\sigma r^2 T^4$<br><br>$3.85 \times 10^{26} = 4\pi \times 5.67 \times 10^{-8} \times r^2 \times 5780^4$ | **C1** |
| | $r = 6.96 \times 10^8$ m | **A1** |

| Question | Answer | Marks |
|---|---|---|
| 10(a)(ii) | $F = L / 4\pi d^2$<br><br>$= (3.85 \times 10^{26}) / (4\pi \times (1.50 \times 10^{11})^2)$ | **C1** |
| | $= 1.36 \times 10^3$ W m$^{-2}$ | **A1** |
| 10(a)(iii) | line of same shape showing peak intensity at greater wavelength | **B1** |
| | line of same shape showing lower peak intensity | **B1** |
| 10(b)(i) | 5 lines in same pattern shifted to longer wavelengths | **B1** |
| 10(b)(ii) | $\Delta\lambda / \lambda = v / c$<br><br>$\Delta\lambda = (21400 / 300000) \times 656$<br>$= 46.8$ nm | **C1** |
| | wavelength $= 656 + 46.8$<br>$= 703$ nm | **A1** |
| 10(b)(iii) | (peak) wavelength too high so temperature too low | **B1** |

| Question | Answer | Marks |
|---|---|---|
| 12(a) | total power of radiation emitted (by the star) | **B1** |
| 12(b) | $F = \frac{L}{4\pi d^2}$<br><br>$= \frac{3.83 \times 10^{26}}{4 \times \pi \times 1.51 \times 10^{112}}$ | **C1** |
| | = 1340 W $m^{-2}$ | **A1** |

| Question | Answer | Marks |
|---|---|---|
| 12(c) | $m = \frac{E}{c^2}$<br><br>$= \frac{3.83 \times 10^{26}}{3.00 \times 10^{82}}$<br><br>$= 4.26 \times 10^{9}$ kg | **A1** |
| 12(d) | $L = 4\pi\sigma r^2 T^4$<br><br>$3.83 \times 10^{26} = 4 \times \pi \times 5.67 \times 10^{-8} \times 6.96 \times 10^{82} \times T^4$ leading to $T = 5770$ K | **B1** |
| 12(e) | $\lambda_{(max)} \propto \frac{1}{T}$<br><br>$\frac{5.00 \times 10^{-7}}{\lambda} = \frac{9940}{5770}$ | **C1** |
| | $\lambda = 2.90 \times 10^{-7}$m | **A1** |

| Question | Answer | Marks |
|---|---|---|
| 10(a) | temperature inversely proportional to wavelength | **M1** |
| | temperature is thermodynamic temperature of surface, and wavelength is the wavelength at which maximum emission rate occurs | **A1** |
| 10(b)(i) | (astronomical) object of known luminosity | **B1** |
| 10(b)(ii) | star / galaxy is moving away from the student | **B1** |
| 10(b)(iii) | one tick placed in correct column in each row:<br>wavelength: too high | **B1** |
| | surface temperature: too low | **B1** |
| | distance: unchanged | **B1** |
| | radius: too high | **B1** |

| Question | Answer | Marks |
|---|---|---|
| 8(a)(i) | movement of star causes change in (observed) frequency<br>**or**<br>movement of star causes redshift | **B1** |
| | observed frequency is lower (than emitted frequency) | **B1** |
| 8(a)(ii) | all three lines shown to left of corresponding printed lines | **B1** |
| | distance between drawn line and corresponding printed line approximately the same for all three lines | **B1** |
| 8(b)(i) | $E = hf$ **and** $\lambda = c / f$ | **C1** |
| | $E = (6.63 \times 10^{-34} \times 3.00 \times 10^{8}) / (488 \times 10^{-9})$<br>$= 4.08 \times 10^{-19}$ J | **A1** |
| 8(b)(ii) | photon energy = $(4.08 \times 10^{-19}) / (1.60 \times 10^{-19})$<br>= 2.55 eV | **C1** |
| | energy level = –3.40 + 2.55<br>= –0.85 eV | **A1** |
| 8(b)(iii) | $\Delta\lambda = \lambda \times (v / c)$<br>$= (488 \times 6.2 \times 10^{6}) / (3.00 \times 10^{8})$<br>( = 10 nm) | **C1** |
| | observed wavelength = 488 + $\Delta\lambda$ = 488 + 10<br>= 498 nm | **A1** |

| Question | Answer | Marks |
|---|---|---|
| 8(c) | $v = H_0 d$ | **C1** |
| | $d = (6.2 \times 10^{6}) / (2.3 \times 10^{-18})$<br>$= 2.7 \times 10^{24}$ m | **A1** |

| Question | Answer | Marks |
|---|---|---|
| 9(a)(i) | speed is (directly) proportional to distance | **M1** |
| | where speed is speed of recession of galaxy (from observer) and distance is distance of galaxy away from observer | **A1** |
| 9(a)(ii) | wavelengths (of spectral lines) are greater (than their known values) | **B1** |
| | redshift shows stars (in distant galaxies) moving away from Earth | **B1** |
| 9(b) | (all) parts of Universe moving away from each other | **B1** |
| | more distant objects are moving away faster | **B1** |
| | matter must have been close together / very dense in the past | **B1** |